

Dear Candidate,

Thank you for your interest in the Internship at Ai Linear.

As part of our interview process, we ask candidates to complete the following engineering challenge. This assignment is intended to simulate a real project that a Systems Engineer is likely to perform while supporting the development, integration, verification, and product qualification of an embedded AI semiconductor platform.

Please spend **about one week** on this assignment. At the end of the week, you will give a **20–30 minute technical presentation** to our engineering team, followed by a technical discussion.

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### Primary Objective (Most Important)

The **primary objective** of this challenge is to **develop a comprehensive verification and validation methodology, together with test software, to thoroughly test the open-source PicoSoC (RV32) microcontroller.**

This assignment is **NOT** primarily about developing firmware for an Mixed Mode System.

This assignment is **NOT** primarily about designing an AI algorithm.

This assignment is **NOT** primarily about designing an Analog Front End.

Instead, your primary responsibility is to determine:

**"How can I thoroughly verify that the PicoSoC microcontroller is capable of reliably serving as the traffic controller for an embedded AI sensing system?"** (the system is described below)

Your presentation should continually focus on this objective.

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### PicoSoC is Configurable

One of the advantages of the open-source PicoSoC project is that it is **configurable**.

The PicoSoC repository contains several optional features and architectural choices that may be enabled, disabled, or modified depending upon the complexity and performance requirements of the target application.

As part of this challenge, we would like you to first evaluate the PicoSoC architecture and determine:

- Which PicoSoC features you would keep.

- Which features you would remove.
- Which features you would add or modify.
- Why your customized PicoSoC is better suited for this application.

Once you have defined your proposed PicoSoC configuration, **your primary task is to develop a verification strategy and test software for your customized PicoSoC implementation.**

In other words,

**First customize PicoSoC.**

**Then thoroughly test your customized PicoSoC.**

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## System Background

Ai Linear is developing an ultra-low-power mixed-signal AI semiconductor platform called **iAcoustic™** for intelligent acoustic and vibration sensing.

One target Mixed Mode System is predictive maintenance, where sounds or vibrations from industrial equipment are analyzed locally to detect anomalies before equipment failure occurs.

For purposes of this assignment, the complete system consists of the following functional blocks.

### 1. Analog Front End (AFE)

The Analog Front End interfaces to microphones or vibration sensors.

For this exercise, you may assume that the AFE periodically provides digitized sensor data to the PicoSoC through a serial interface. The PicoSOC must give priority to this interface because it is the source of real time data.

You do **not** need to design or understand the internal analog circuitry.

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### 2. PicoSoC (RV32)

The PicoSoC serves as the **central embedded traffic controller** for the entire system.

It coordinates communication among all remaining system blocks. Control and data are transmitted among the blocks on a 3 wire Serial Peripheral Interface (SPI).

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### 3. SPI EEPROM

The EEPROM stores:

- System configuration information
- Machine Learning training coefficients
- Calibration information
- Other system parameters

The Machine Learning training coefficients are generated offline using Ai Linear's proprietary software called **iADE**.

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### 4. AI Processing Engine (Assume an ARM-Based Processing Engine)

Because the architecture of Ai Linear's proprietary iAcoustic AI accelerator cannot be disclosed, please assume that this block is an **ARM-based processing engine** connected to the PicoSoC via the SPI.

The PicoSoC is responsible for:

- Initializing the AI engine
- Reading the ML training coefficients from the EEPROM
- Downloading the coefficients into the AI processing engine
- Writing configuration registers
- Reading status registers
- Monitoring interrupt/status flags & timely responding appropriately
- Sending digitized sensor data from the AFE
- Receiving anomaly detection results

The internal operation of the AI processing engine is **outside the scope** of this assignment.

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### 5. Wireless Transmitter

When the AI engine detects an anomaly, the PicoSoC sends the anomaly report and system status to the wireless transmitter, which then communicates with an external receiver.

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## Overall System Operation

The overall system operates approximately as follows:

1. Ai Linear's iADE software trains a Machine Learning model using datasets containing normal and abnormal acoustic or vibration signals.
2. The resulting Machine Learning training coefficients are stored in the SPI EEPROM.
3. During system startup, the PicoSoC reads these coefficients from the EEPROM and downloads them into the AI Processing Engine.
4. During normal operation, the Analog Front End continuously provides digitized sensor data.
5. The PicoSoC transfers this sensor data to the AI Processing Engine.
6. The AI Processing Engine analyzes the incoming data and determines whether an anomaly is present.
7. If an anomaly is detected, the PicoSoC retrieves the result and forwards it to the wireless transmitter.

Again, **the primary objective of this assignment is to verify that the PicoSoC can reliably manage all of this system traffic under every realistic operating condition.**

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## Reference Material

Please review the following references.

### PicoRV32 / PicoSoC Repository

<https://github.com/YosysHQ/picorv32>

### Microchip SPI EEPROM Datasheet

<https://ww1.microchip.com/downloads/en/DeviceDoc/22040A.pdf>

### RISC-V Specifications

<https://riscv.org/technical/specifications/>

### SPI Protocol Reference

[https://en.wikipedia.org/wiki/Serial\\_Peripheral\\_Interface](https://en.wikipedia.org/wiki/Serial_Peripheral_Interface)

## **Representative MEMS Evaluation Board**

Same Sky DEVKIT-MEMS-001

<https://www.sameskydevices.com/product/evaluation-board/devkit-mems-001>

## **Representative SPI-to-USB Adapter**

Adafruit FT232H Breakout

<https://www.adafruit.com/product/2264>

These hardware references are provided only to give context for the Analog Front End. You are **not** expected to purchase or build any hardware.

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## **Assignment**

Please prepare a presentation covering the following topics.

### **1. PicoSoC Evaluation**

Explain your understanding of the PicoSoC architecture.

Describe:

- Which PicoSoC options you would enable.
  - Which options you would disable.
  - Any modifications you recommend.
  - Why your customized PicoSoC is appropriate for this application.
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### **2. System Block Diagram**

Prepare one or more block diagrams showing communication among:

- Analog Front End
- PicoSoC
- SPI EEPROM
- AI Processing Engine
- Wireless Transmitter

Clearly identify all major data flows.

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### **3. Verification Strategy (Primary Focus)**

Develop a comprehensive verification methodology for your customized PicoSoC.

Examples include:

- Power-up initialization
- EEPROM read/write verification
- Downloading ML coefficients into the AI engine
- Continuous sensor acquisition
- Simultaneous sensor acquisition and EEPROM access
- Simultaneous wireless transmission and sensor acquisition
- SPI bus contention
- Interrupt handling and collisions
- Wait-state analysis
- Buffer overflow and underflow
- Timeout handling
- Communication failures
- Startup and recovery
- Reset recovery
- Brown-out conditions
- Sleep and wake-up transitions
- Power management
- Worst-case traffic conditions
- Long-duration stress testing

Please include any additional verification scenarios that you believe are important.

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#### **4. Test Software (Primary Focus)**

Describe and, if time permits, implement software that thoroughly exercises your customized PicoSoC.

Examples include:

- Automated regression testing
- Unit testing
- Integration testing
- Random traffic generation
- Fault injection
- SPI protocol verification
- Traffic simulation
- Functional coverage
- State-machine coverage
- Code coverage

Your goal is to maximize confidence that the PicoSoC has been thoroughly verified.

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#### **5. Deliverables**

Please prepare:

- PowerPoint or PDF presentation
- System block diagrams
- Verification architecture
- Verification plan
- Test matrix
- State-machine diagrams
- Source code (if developed)
- Simulation results (if available)

- Discussion of verification coverage
  - List of engineering assumptions
  - Questions you would ask before product qualification
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## Evaluation

We understand that this challenge is broader than can typically be completed within one week.

We **do not** expect a complete implementation.

We are primarily interested in evaluating:

- Your engineering thought process
- Your understanding of embedded systems
- Your ability to customize an existing open-source platform for a specific application
- Your verification methodology
- The quality and completeness of your proposed test software
- Your software engineering skills
- Your communication and presentation skills

The strongest submissions will demonstrate a structured engineering approach, identify additional verification scenarios beyond those listed above, and propose practical methods to maximize confidence that the PicoSoC is fully qualified to serve as the embedded traffic controller within the iAcoustic platform.

We look forward to discussing your work and learning more about your engineering approach.

Best regards,